

ADI Internship

LAB

Audio Spectrum Analyzer SoC

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SoC Architecture:

The proposed System on Chip (SoC) architecture is centered around a microcontroller core interfaced with dedicated hardware accelerators over a standard system bus. Key components include:

- **Communication Interface:** An SPI Slave peripheral is Used to receive audio samples and transmit the peak frequency back to the host PC.
- **Memory Storage:** A single-port RAM block to store the incoming audio samples before processing.
- **Hardware Accelerator:** A dedicated FFT processing block to offload the heavy computational burden from the main processor.
- **Direct Memory Access (DMA):** To move data between the SPI peripheral, the single-port RAM, and the FFT accelerator efficiently.

System Block Diagram:

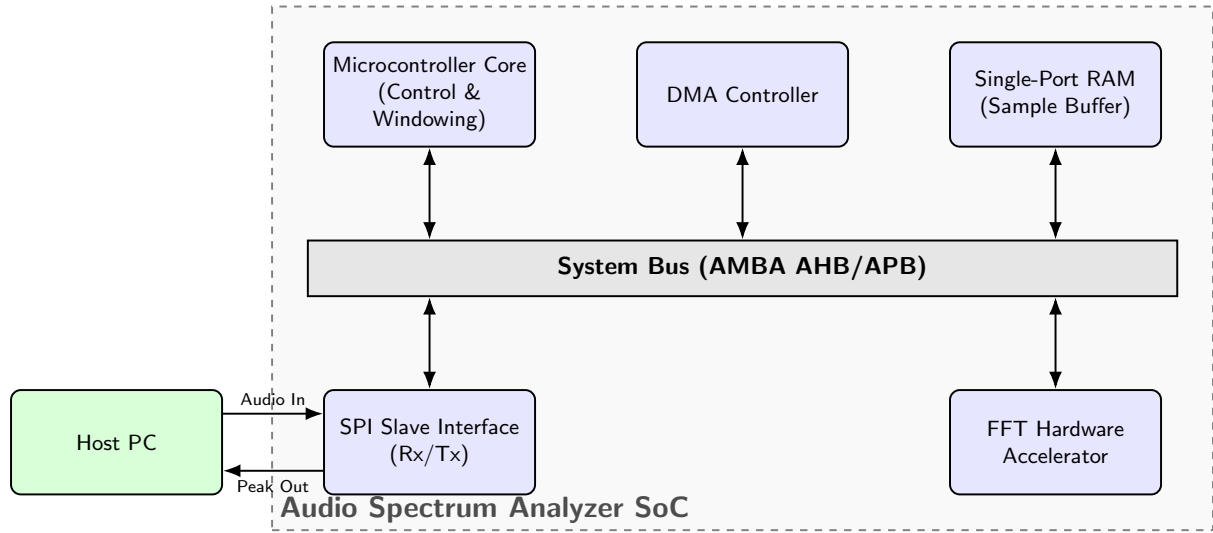


Figure 1: Hardware/Software Co-design SoC Block Diagram

Hardware/Software Partitioning with Reasoning:

Hardware Allocation

- **SPI Slave & Memory Controller:** Implemented in digital hardware to handle the strict timing requirements of serial communication and rapid memory access.
- **FFT Computation:** The FFT algorithm is computationally intensive; Implementing as a hardware structure significantly accelerates processing time and reduces power consumption compared to software execution.
- **Magnitude Spectrum :** Squaring and adding the real and imaginary parts ($I^2 + Q^2$) can be pipelined efficiently within the hardware accelerator.

Software Allocation

- **System Control:** The main application running on the processor determines the data flow, configures the DMA, and handles interrupts from the hardware blocks.

- **Windowing:** Can be done in software by multiplying incoming samples with pre-calculated window coefficients stored in memory. This offers the flexibility to easily switch between different window types depending on the specific frequency resolution needs without altering the hardware.
- **Peak Finding and Conversion:** Iterating through an array of magnitude values to find the maximum bin and multiplying by the frequency resolution factor is a highly sequential task that is well-suited for a simple software loop.

System Application Pseudocode

The following pseudocode outlines the software control loop running on the main processor.

```
// SoC Audio Spectrum Analyzer Main Application

#define FFT_SIZE 1024
#define SAMPLE_RATE 20480
#define FREQ_RES (SAMPLE_RATE / FFT_SIZE)

uint16 audio_buffer[FFT_SIZE];
uint32 mag_buffer[FFT_SIZE/2];

void main() {
    // Initializing System Peripherals
    SPI_Slave_Init();
    DMA_Init();
    FFT_Accelerator_Init();

    while(1) {
        // Receive samples
        // Wait for SPI DMA to fill the single-port RAM buffer
        WaitForInterrupt(DMA_RX_COMPLETE);

        // Windowing
        ApplyWindowFunction(audio_buffer, FFT_SIZE);

        // Compute FFT & Magnitude
        // Trigger HW Accelerator to read from RAM, compute, and output
        FFT_Start(audio_buffer, mag_buffer);
        WaitForInterrupt(FFT_COMPLETE);

        // Find Peak Bin
        int peak_bin = 0;
        uint32 max_mag = 0;
        for(int i = 0; i < FFT_SIZE/2; i++) {
            if(mag_buffer[i] > max_mag) {
                max_mag = mag_buffer[i];
                peak_bin = i;
            }
        }

        // Convert to Frequency
        uint32 peak_freq = peak_bin * FREQ_RES;

        // Send result back to Host PC via SPI
        SPI_Transmit(peak_freq);
    }
}
```